

Brief summary of this function: Plots a continuous section of the charging cycle that is of interest

Detailed explanation of this function: This helper function goes hand in hand with the `plotcycle_modes` one as it is used by that function to plot one charging phase at a time. Its purpose is find all the points apart of one phase (ie: CV) and if the selected phase does not show in the data, the function will stop running. Then, this functions checks for any gaps that are in the data to see where one continuous section ends and where the next begins. Each of the continuous sections are plotted separately using the determined colors in `plotcycle_modes` so that the unrelated sections are not connected.

```
%need to start with function to establish as helper function

function line = plotMode(time, y, mode, mode_name, color) %different factors
in the function

line = [];

    % used to find all indices of the selected mode
    idx = find(mode == mode_name); %makes sure that mode = mode name (ie:
CC,CV) and finds where it appears

    %making sure that if there is nothing to find it stops the function
    %(doesn't keep going on)
    if isempty(idx)
        return
    end

    start = 1; %where the first section starts

    %now need to find the breaks in the mode
    for i = 1:length(idx)-1

        if idx(i+1)-idx(i)>1 %plot the current section if there is a gap

            segment = idx(start:i);

            %the first three parts of this graph will be in the second
            %helper function (CycleModes)
            p = plot(time(segment), y(segment), 'Color', color, 'LineWidth',
1.5);

            if isempty(line)
                line = p;
            end

            %starting the new section
            start = i+1;
        end
    end
end
```

```
%making sure that the final section is plotted from start-end
segment = idx(start:end);

%once again the first three parts will be expanded upon in CycleModes
p = plot(time(segment), y(segment), 'Color', color, 'LineWidth', 1.5);
if isempty(line)
    line = p;
end
end
```